

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 5

HINTS

We will discuss several questions from Chapter 5 in Ross' book.

Tutorial Questions

- (1) Consider n coins, each of which independently comes up heads with probability p . Suppose that n is large and p is small, and let $\lambda = np$. Suppose that all n coins are tossed; if at least one comes up heads, the experiment ends; if not, we again toss all n coins, and so on. That is, we stop the first time that at least one of the n coins come up heads.

- Let X denote the total number of heads that appear.

Which of the following reasonings concerned with approximating $P\{X = 1\}$ is correct (in all cases, Y is a Poisson random variable with parameter λ)?

- (a) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ ,

$$P\{X = 1\} \approx P\{Y = 1\} = \lambda e^{-\lambda}.$$

- (b) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ , and because we stop only when this number is positive,

$$P\{X = 1\} \approx P\{Y = 1 | Y > 0\} = \frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}}.$$

- (c) Because at least one coin comes up heads, X will equal 1 if none of the other $n - 1$ coins comes up heads. Because the number of heads resulting from these $n - 1$ coins is approximately Poisson with mean $(n - 1)p \approx \lambda$,

$$P\{X = 1\} \approx P\{Y = 0\} = e^{-\lambda}.$$

Solution: Approach (b) is correct. This can also be verified using geometric distribution as

$$P\{X = 1\} \approx \sum_{i=0}^{\infty} (P\{Y = 0\})^i P\{Y = 1\} = \frac{P\{Y = 1\}}{1 - P\{Y = 0\}}.$$

Approach (a) is wrong because it does not consider the fact that there could be multiple failed rounds before the experiment ends – but the unconditional probability $P\{X = 1\}$ takes this into account. Hence the true probability will be larger than the single-round probability $P\{Y = 1\}$.

Approach (c) is wrong because it does not consider multiple rounds and, for the unconditional probability $P\{X = 1\}$ one needs to enforce that one of the coins must be head. Simply considering there are $n - 1$ tails is not enough – the remaining coin could be either head or tail.

(2) Consider the function

$$f(x) = \begin{cases} C(2x - x^3), & 0 < x < \frac{5}{2} \\ 0, & \text{otherwise} \end{cases}$$

Could f be a probability density function? If so, determine C . How about $g(x)$ given by

$$g(x) = \begin{cases} C(2x - x^2), & 0 < x < \frac{5}{2} \\ 0, & \text{otherwise} \end{cases}$$

Answer: (a) If f is a probability density function, then we should have (1) $f(x) \geq 0$ for all x and (2) $1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^{\frac{5}{2}} C(2x - x^3)dx = \left[C \left(x^2 - \frac{1}{4}x^4 \right) \right]_0^{\frac{5}{2}} = C \left(\frac{25}{4} - \frac{1}{4} \cdot \frac{625}{16} \right) = C \left(-\frac{225}{64} \right)$, which gives $C = -\frac{64}{225}$. Now, for $x = 1$, $f(1) = -\frac{64}{225}(2 \times 1 - 1^3) = -\frac{64}{225} < 0$, and (1) cannot be satisfied. This shows that f given cannot a probability density function.

(b) The same argument applies to this question. Note that if f in this question is a density function, then $C = -\frac{24}{25}$ and now try $f(2.1)$, which is negative.

(3) The probability density function of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

If $\mathbb{E}(X) = 3/5$, find a and b .

Answer: First, we have

$$\begin{aligned} \frac{3}{5} = \mathbb{E}(X) &= \int_0^1 xf(x)dx = \frac{a}{2} + \frac{b}{4}, \\ 2a + b &= \frac{12}{5}. \end{aligned}$$

Also we have

$$\begin{aligned} 1 &= \int_0^1 f(x)dx = a + \frac{b}{3}, \\ 3a + b &= 3. \end{aligned}$$

We thus have $a = 3/5$, $b = 6/5$.

(4) The probability density function of X , the lifetime of a certain type of electronic device (measured in hours), is given by

$$f(x) = \begin{cases} \frac{10}{x^2}, & \text{if } x > 10; \\ 0, & \text{if } x \leq 10. \end{cases}$$

(a) Find $P\{X > 20\}$.

(b) What is the cumulative distribution function of X ?

(c) What is the probability that of 6 such types of devices at least 3 will function for at least 15 hours? What assumptions are you making?

Answer:

(a)

$$P\{X > 20\} = \int_{20}^{\infty} \frac{10}{x^2} dx = \frac{1}{2}.$$

(b) When $x \geq 10$,

$$P\{X \leq x\} = \int_{-\infty}^{10} 0 dt + \int_{10}^x \frac{10}{t^2} dt = 0 + \left[10 \left(-\frac{1}{t} \right) \right]_{10}^x = 1 - \frac{10}{x}.$$

Hence the CDF of X is

$$F(x) = \begin{cases} 1 - \frac{10}{x}, & x > 10; \\ 0, & x \leq 10. \end{cases}$$

(c) Assuming the independence of the devices. Each device works for at least 15 hours with probability $1 - F(15) = 2/3$, hence the probability in question is

$$1 - \left(\frac{1}{3}\right)^6 - \binom{6}{1} \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^5 - \binom{6}{2} \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^4 = \frac{656}{729} \approx 0.8999$$

(5) Let $X \sim N(3, 16)$. Find $P\{X > 0\}$.

Answer: For this distribution, $\mu = 3$ and $\sigma = 4$. Let $Z = (X - 3)/4$. Z is a standard normal variable, and

$$P\{X > 0\} = P\{(X - 3)/4 > (0 - 3)/4\} = P\{Z > -0.75\} = P\{Z < 0.75\} \approx 0.7734.$$

(6) Stanford accepts 2480 students and each student has a 68% chance of attending. Let X be the number of students who will attend. Then $X \sim \text{Bin}(2480, 0.68)$. What is $P(X > 1745)$?

Answer: We first calculate $E(X) = 2480 \times 0.68 = 1686.4$ and $\text{Var}(X) = 2480 \times 0.68 \times (1 - 0.68) = 539.7$. Thus $\sigma = 23.23$.

We now use $Y \sim N(1686.4, 539.7)$ to approximate X , and we have

$$\begin{aligned} P\{X > 1745\} &\approx P\{Y > 1745.5\} = P\left\{ \frac{Y - 1686.4}{23.23} > \frac{1745.5 - 1686.4}{23.23} \right\} \\ &= 1 - \Phi(2.54) = 0.0055. \end{aligned}$$

(7) If X is an exponential random variable with parameter λ , and $c > 0$, show that cX is exponential with parameter λ/c .

Answer: Let $Y = cX$, then

$$P\{Y \geq y\} = P\left\{X \geq \frac{y}{c}\right\} = e^{-(\lambda/c)y},$$

which implies that Y is exponential with parameter λ/c .

Students ask the question of how to obtain $P\{Y \geq y\} = P\left\{X \geq \frac{y}{c}\right\}$. Note that $\{Y \geq y\}$ is just the $\{\omega \in \Omega : Y(\omega) \geq y\}$, where Y is a random variable and y is a real number (recall that a random variable Y is a function from the same space Ω to the real numbers). By $Y = cX$, we have

$$\begin{aligned} P\{Y \geq y\} &= \{\omega \in \Omega : Y(\omega) \geq y\} = \{\omega \in \Omega : (cX)(\omega) \geq y\} \\ &= \{\omega \in \Omega : c(X(\omega)) \geq y\} = \{\omega \in \Omega : X(\omega) \geq y/c\} = P\left\{X \geq \frac{y}{c}\right\}. \end{aligned}$$

Assignment Questions

- (1) If Y is uniformly distributed over $(0, 5)$, what is the probability that the roots of the equation $4x^2 + 4xY + Y + 2 = 0$ are both real?

Answer: The equation has real roots if and only if

$$16Y^2 - 16(Y + 2) \geq 0 \Leftrightarrow Y^2 - Y - 2 \geq 0 \Leftrightarrow Y \leq -1 \text{ or } Y \geq 2.$$

Since Y is uniform on $(0, 5)$, and $f_Y(x) = \frac{1}{5}$ for $x \in (0, 5)$, and 0 for $x \notin (0, 5)$. $P(Y \leq -1) = 0$. Thus

$$P(Y \leq -1 \text{ or } Y \geq 2) = P(Y \leq -1) + P(Y \geq 2) = \int_{-\infty}^{-1} 0 dx + \int_2^5 \frac{1}{5} dx = 0 + \frac{3}{5} = \frac{3}{5}.$$

- (2) One thousand independent rolls of a fair die will be made. Compute an approximation to the probability that the number 6 will appear between 150 and 200 times inclusively. If the number 6 appears exactly 200 times, find the probability that the number 5 will appear less than 150 times.

Answer: Let X denote the number that 6 appears. Then X follows the binomial distribution $\text{Bin}(1000, 1/6)$, and thus $\mathbb{E}(X) = 1000 \times 1/6 = 1000/6$ and $\text{Var}(X) = 1000/6 \times (1 - 1/6) = 5000/36$. Therefore, to approximate this binomial distribution by normal distribution, we let

$$Z = \frac{X - \frac{1000}{6}}{\sqrt{\frac{5000}{36}}},$$

we have

$$\begin{aligned} P\{150 \leq X \leq 200\} &= P\{149.5 \leq X \leq 200.5\} \\ &= P_{Z \sim N(0,1)} \left\{ \frac{149.5 - \frac{1000}{6}}{\sqrt{\frac{5000}{36}}} \leq Z \leq \frac{200.5 - \frac{1000}{6}}{\sqrt{\frac{5000}{36}}} \right\} \\ &\approx P_{Z \sim N(0,1)} \{-1.46 \leq Z \leq 2.87\} \\ &= \Phi(2.87) - (1 - \Phi(1.46)) \\ &\approx 0.9979 - (1 - 0.9279) \\ &= 0.9258. \end{aligned}$$

Given that the number 6 appears 200 times, the number Y that 5 appears follows the binomial distribution $\text{Bin}(800, 1/5)$, so we have $1000 - 200 = 800$ rolls not containing 6, and for each such a roll, as we already know that number 6 cannot appear, the probability that the number 5 appears is $1/5$. Thus, $\mathbb{E}(Y) = 800 \times 1/5 = 160$ and $\text{Var}(Y) = 800 \times 1/5 \times (1 - 1/5) = 128$. This gives

$$\begin{aligned} P\{Y < 150\} &= P\{Y \leq 149.5\} \\ &\approx P_{Z \sim N(0,1)} \left\{ Z \leq \frac{149.5 - 160}{\sqrt{128}} \right\} \\ &\approx P_{Z \sim N(0,1)} \{Z \leq -0.93\} \\ &= 1 - \Phi(0.93) \\ &\approx 1 - 0.8238 \\ &= 0.1762. \end{aligned}$$

(Remark: The exact answers are $0.922995 \dots$ and $0.176984 \dots$).

- (3) Suppose that X is a normal random variable with mean 5. If $P\{X > 9\} = 0.2$, approximately what is $\text{Var}(X)$?

Answer: $P\{X < 9\} = 1 - P\{X > 9\} = 1 - 0.2 = 0.8$. Suppose that the variance is σ^2 . We first transform X to a standard normal variable by letting $Z = \frac{X - 5}{\sigma}$, and then

$$P\{X < 9\} = P_{Z \sim N(0,1)} \left\{ Z < \frac{9 - 5}{\sigma} \right\} = 0.8,$$

and then look at the table for cumulative normal distribution, find the value closest to 0.8, which is 0.84 and this gives

$$\frac{9 - 5}{\sigma} \approx 0.84, \\ \sigma^2 \approx \left(\frac{4}{0.84} \right)^2 \approx 22.675.$$

- (4) Let Z be a standard normal random variable, and let g be a differentiable function with derivative g' .
- (a) Show that $\mathbb{E}(g'(Z)) = \mathbb{E}(Zg(Z))$, assuming that $g(x)/e^{x^2/2} \rightarrow 0$ as $|x| \rightarrow \infty$.
 - (b) Show that $\mathbb{E}(Z^{n+1}) = n \mathbb{E}(Z^{n-1})$.
 - (c) Find $\mathbb{E}(Z^4)$.

Answer:

$$\begin{aligned} & \mathbb{E}(g'(Z)) \\ &= \int_{-\infty}^{\infty} g'(z) \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \left[g(z) \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \right]_{z=-\infty}^{\infty} - \int_{-\infty}^{\infty} g(z) \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} (-z^2/2)' dz \quad (\text{integration by parts}) \\ &= 0 + \int_{-\infty}^{\infty} z g(z) \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \mathbb{E}(Zg(Z)). \end{aligned}$$

- (b) This part follows from part (a) by setting $g(z) = z^n$, we have

$$\mathbb{E}(Z \cdot Z^n) = \mathbb{E}((Z^n)') = \mathbb{E}(nZ^{n-1}) = n \mathbb{E}(Z^{n-1}).$$

- (c) Invoking part (b), $\mathbb{E}(Z^4) = 3 \cdot \mathbb{E}(Z^2) = 3 \cdot 1 \cdot \mathbb{E}(Z^0) = 3$.

We can also use the fact that $\mathbb{E}(Z^2) = (\mathbb{E}(Z))^2 + \text{Var}(Z) = 0^2 + 1 = 1$ to get the last equality. Remember that Z is a standard normal variable and $\mathbb{E}(Z) = 0$ and $\sigma = 1$.

- (5) Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait
- (a) more than 10 minutes;
 - (b) between 10 and 20 minutes.

Answer: Let X denote the length of the call made by the person in the booth. Then the desired probabilities are

- (a) $P\{X > 10\} = 1 - F(10) = e^{-1} \approx 0.368$.
- (b) $P\{10 < X < 20\} = F(20) - F(10) = e^{-1} - e^{-2} \approx 0.233$.